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Plasma metagenomic cfDNA sequencing identifies pathogens in culture-negative sepsis following urinary pouch rupture

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SUMMARY

A patient with a complex urological history presented with abdominal pain and respiratory distress after catheter dysfunction. She underwent emergency surgery for a ruptured urinary pouch. Sepsis was later diagnosed based on clinical deterioration, including tachycardia, fever, an elevated respiratory rate and raised inflammatory markers, but blood cultures remained negative. A metagenomic microbial cell-free DNA (cfDNA) assay (ISEP-SEQ), performed early from plasma as part of a research protocol, identified *Klebsiella* and *Pseudomonas* at the genus level. Results were obtained retrospectively and were not available in real time; therefore, they did not alter immediate management. These findings were confirmed by cultures from drainage fluid and urine. Broad-spectrum antibiotic treatment led to clinical improvement. This case highlights the limitations of conventional microbiological methods in culture-negative sepsis and illustrates the role of cfDNA-based metagenomic testing as an adjunctive and complementary diagnostic tool for early, accurate pathogen detection. Early use of such tools may support timely and targeted management in complex infectious disease presentations.

BACKGROUND

Sepsis is a life-threatening syndrome caused by the immune system's dysregulated response to infection, leading to organ damage, septic shock, and death. The condition is characterised by an increase of two or more points in the Sequential Organ Failure Assessment (SOFA) score.¹ Globally, sepsis accounts for approximately 20% of deaths, as highlighted by the WHO Sepsis Report 2017.²

Current guidelines recommend immediate empirical broad-spectrum antibiotic therapy following blood culture collection, with transition to pathogen-specific de-escalation based on microbiological results.¹

However, conventional culture techniques often require 2–7 days for pathogen identification³ and may yield false negatives in 40%–60% of cases with suspected sepsis, particularly in patients with prior antibiotic exposure or complex infections.^{3–5} This hampers the initiation of targeted therapy and contributes to the growing threat of antimicrobial resistance (AMR).

Advances in molecular diagnostics, such as plasma metagenomics microbial cell-free DNA (mcfDNA)

sequencing, offer a novel approach to early pathogen detection. This 'liquid biopsy' technology can identify a broad range of pathogens directly from blood samples, including atypical organisms, and may provide clinically actionable results even when standard cultures are negative.^{6,7}

This case illustrates its potential clinical utility.

CASE PRESENTATION

A woman in her early 30s under regular urological follow-up presented directly to a urology outpatient clinic of a tertiary care hospital in Germany, with severe lower abdominal pain and progressive dyspnoea following catheter dysfunction. Her relevant medical history included a lumbosacral meningocele, which required multiple prior neurosurgical procedures, including spinal untethering, placement of a ventriculoperitoneal shunt and posterior spinal stabilisation. She had also developed low bladder compliance, which was managed with ileocaecal augmentation and an appendix stoma.

During follow-up, she developed transitional cell carcinoma of the augmented bladder, which progressed to keratinising squamous cell carcinoma. This was managed with radical cystectomy and conversion to a continent urinary diversion (MAINZ pouch I, a continent ileocaecal urinary reservoir with catheterisable stoma), with preservation of the appendix stoma. She was monitored through regular oncological evaluations and managed bladder emptying via clean intermittent catheterisation.

In the weeks prior to admission, the patient reported increasing difficulty with catheterisation. A pouchoscopy revealed mucosal swelling at the stoma site, and a catheter was placed for continuous drainage for 3 weeks without immediate complications.

The night before presentation, catheter function deteriorated, with reduced urine output followed by acute respiratory symptoms. On admission, she showed signs of cardiopulmonary decompensation, and emergency computer tomography (CT) demonstrated a large rupture of the urinary reservoir. Emergency lower abdominal laparotomy confirmed an iatrogenic pouch rupture, which was surgically repaired. A catheter fixation device (Pouchofix, Teleflex)



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and a nasogastric tube were placed. Relevant laboratory findings on admission and throughout her hospital stay are presented in [table 1](#).

Postoperatively, the patient was transferred to the intensive care unit (ICU). On admission, she was alert (Glasgow Coma Scale score 14), with pupils equal and reactive, tachycardia (120 beats per minute), systolic blood pressure of 110 mm Hg, and oxygen saturation of 98% on 3 L/min oxygen via nasal cannula. The abdomen was soft and moderately distended without peritonism; surgical drains showed minor haemorrhagic output. Piperacillin–tazobactam was initiated intraoperatively and continued postoperatively.

On Day 2 of ICU admission, due to ongoing clinical instability, antimicrobial therapy was escalated to meropenem and fluconazole. Shortly after administration, the patient developed pruritus, flushing, wheeze and agitation, consistent with an acute hypersensitivity reaction; therapy was

discontinued and managed with antihistamines (dimetindene, cimetidine) and corticosteroids (methylprednisolone), with rapid symptom improvement. Reintroduction of antimicrobial therapy triggered a second hypersensitivity reaction, prompting a switch to ceftriaxone, which was well tolerated and continued as definitive therapy. The subsequent ICU course (Days 2–9) was characterised by persistent tachycardia, fluctuating infection markers, evolving pulmonary findings, and negative blood cultures. These events and the laboratory parameter course are summarised in [table 2](#).

A metagenomic microbial cfDNA assay performed on a blood sample collected on Day 3 as part of a clinical research protocol (approved by the regional ethics committee; study ID 2020–15535) identified *Klebsiella* and *Pseudomonas* species. These pathogens were later confirmed by cultures from drainage fluid and urine obtained on Days 8 and 9, respectively. At the time of culture confirmation, the patient was clinically improving and was transferred from the ICU to the urology monitoring ward in stable condition ([figure 1](#)).

Definitions box

Sepsis-3

- Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection.
- Operationally, sepsis is identified by an acute increase of ≥ 2 points in the Sequential Organ Failure Assessment score, as evidence of organ failure.

Criteria for septic shock (per Sepsis-3)

Septic shock is defined by both:

- (1) Persistent hypotension requiring vasopressors to maintain a mean arterial pressure ≥ 65 mm Hg;
- (2) Serum lactate > 2 mmol/L (> 18 mg/dL), despite adequate fluid resuscitation.

Culture-negative sepsis (per Sepsis-3)

- Clinical diagnosis of sepsis with negative conventional cultures, despite appropriate sampling (blood, urine, respiratory samples, etc.) within a reasonable diagnostic timeframe.
- Occurs in up to 40%–60% of suspected cases, often due to prior antibiotics, difficult sampling (e.g., altered anatomy), low pathogen burden, or infection with fastidious/unculturable organisms.

Plasma metagenomic cfDNA

- Plasma metagenomic cell-free DNA (cfDNA) sequencing is a non-invasive test that detects microbial cfDNA fragments in plasma using next-generation sequencing.
- It can identify bacteria, fungi, DNA viruses, and some parasites at genus or species level, and reports relative abundance metrics (e.g., reads per million).
- Some platforms can also report antimicrobial resistance markers.
- Typical turnaround is 24–72 hours after laboratory receipt.

Limitations and pitfalls of cfDNA sequencing

- Does not distinguish viable from non-viable organisms.
- May reflect contamination or transient DNAemia.
- Limited detection of RNA viruses or pathogens with very low cfDNA shedding.
- Does not provide infection source, host response severity, or functional virulence data.

INVESTIGATIONS

Imaging studies

Contrast-enhanced abdominal CT was performed on admission. During the excretory phase, following retrograde contrast instillation via an indwelling pouch catheter, significant contrast leakage into the abdominal cavity was identified, originating from a defect in the left laterodorsal aspect of the urinary pouch. Free intraperitoneal fluid was detected in the perisplenic, perihepatic and pelvic spaces.

CT of the chest excluded pulmonary embolism and right heart strain.

Abdominal ultrasound confirmed a small volume of free fluid in the pelvic cavity (Koller pouch) with no further abnormalities.

Supine chest radiography demonstrated an enlarged cardiac silhouette and a left lower lobe consolidation with a positive bronchopneumogram, consistent with pneumonia, as well as a right-sided pleural effusion.

Transthoracic echocardiography showed a volume-depleted state with kissing ventricles and a collapsible inferior vena cava, supporting the need for aggressive fluid management.

Continuous physiological monitoring was maintained throughout the patient's ICU stay, including 5-lead electrocardiography, invasive and non-invasive blood pressure, respiratory rate and body temperature. These parameters are summarised in [figures 2 and 3](#).

Microbiological investigations

Two pairs of blood cultures obtained on Day 2 (using BD BACTEC Standard/10 Aerobic/F and Standard/Anaerobic/F bottles (BD Becton Dickinson, USA); and incubated using the BD BACTEC FX40 system) remained negative after 5 days of incubation.

Drainage fluid from the surgical site (Easy Flow drain) was cultured repeatedly

Initial Sample (Day 2)

After 4 days of incubation, *Klebsiella oxytoca* was isolated in low quantities. Identification was confirmed using the MALDI Biotyper (Bruker Daltonics), and susceptibility testing was performed with the VITEK2 system (bioMérieux) using the VGN-14 card. Results were interpreted according to European Committee on Antimicrobial Susceptibility Testing (EUCAST)

Table 1 Trends in laboratory parameters throughout ICU treatment

	Sampling day								
	Unit	Day 1 pre-op	Day 1 post-op	Day 2	Day 3	Day 4	Day 5	Day 6	Normal range
Clinical chemical parameters									
Sodium	mmol/L	138	144	139	145	142	137	137	136–145
Potassium	mmol/L	3.7	4.2	3.9	4.6	3.8	3.8	4.3	3.4–4.9
Chloride	mmol/L	105	155	109	122	105	100	103	98–107
Calcium	mmol/L	2.4	2.14	2.27	2.18	2.14	2.21	2.23	2.15–2.58
Inorganic phosphorus	mg/dL	3.1	2.6	2.9	2.8	2.8	3.8	2.7	2.3–4.7
Creatinine	mg/dL	1.08	0.39	0.34	0.36	0.38	0.33	0.31	0.55–1.02
eGFR	mL/min/1.73 m ²	68	139	145	143	140	147	150	77–129
Urea	mg/dL	19	7	7	11	10	6	5	7–19
CRP	mg/L	1.7	68	361	402	336	363	378	<5
PCT	ng/mL			4.7	3.2	1.9	0.78		<0.5
IL-6	pg/mL				117				<7
Basic haematological parameters									
WBC	x10 ⁹ /L	15.3	3.6	10.0	13.9	10.2	10.8	13.1	3.5–10
RBC	x10 ¹² /L	4.91	3.69	3.20	3.24	3.20	3.36	3.34	3.7–4.8
HGB	g/L	143	109	95	95	93	99	97	120–160
HCT	L/L	0.427	0.326	0.284	0.282	0.286	0.298	0.293	0.34–0.44
PLT	x10 ⁹ /L	513	265	249	271	296	281	290	150–360
Blood gas analysis									
pH			7.42 (art)	7.46 (art)	7.42 (ven)	7.43 (ven)	7.38 (ven)	7.43 (ven)	7.35–7.45
pCO ₂	mmHg		30	35	43	46	45	43	32–45
pO ₂	mmHg		100	71	42	43	41	36	21.8–26.2
cHCO ₃	mmol/L		21.1	25.4	26.8	28.8	24.9	26.9	21.8–26.2
BE	mmol/L		−4.0	1.1	3.0	5.3	1.0	3.4	−2–3
sO ₂	%		98.5	95.5	76.1	75.8	72.2	67.0	95–99
Lactate	mmol/L		4.7	3.2	0.6	0.9	0.6	0.5	0.5–1.6

Averaged values in 8-hour intervals.

BE, base excess; cHCO₃, bicarbonate concentration; CRP, C-reactive protein; eGFR, estimated glomerular filtration rate; HCT, haematocrit; HGB, haemoglobin; HR, heart rate; ICU, intensive care unit; IL-6, interleukin-6; MAP, mean arterial pressure; mg, milligram; mL/min/1.73 m², millilitres per minute per 1.73 square metres of body surface area; mm Hg, millimetres of mercury; mmol, millimole; ng, nanogram; nL, nanolitre; pCO₂, partial pressure of carbon dioxide; PCT, procalcitonin; pg, picogram; pH, measure of acidity/alkalinity; pL, picolitre; PLT, platelet; pO₂, partial pressure of oxygen; RBC, red blood cell; RR, respiratory rate; sO₂, oxygen saturation; WBC, white blood cell.

guidelines, showing susceptibility to most antibiotics except ampicillin.

Follow-up Sample (Day 5)

No microbial growth.

Repeated Sample (Day 7):

Vancomycin-resistant *Enterococcus faecium* was isolated in low quantities after 5 days. Identification and susceptibility profiling were performed using the MALDI Biotyper and VITEK®2 (AST-P655 card). The isolate was resistant to ampicillin, ampicillin/sulbactam, imipenem, vancomycin and levofloxacin, but susceptible to teicoplanin, linezolid and tigecycline. PCR confirmed the presence of the vanB resistance gene.

Urine cultures

Conduit urine (Day 2)

Cultures were negative for bacterial growth.

Catheterised urine (Day 7)

No microbial growth was observed.

Pouch urine sample (Day 7):

Pseudomonas aeruginosa was identified at a low concentration (10² CFU/mL). The isolate was susceptible to meropenem, tobramycin and amikacin, dose-dependent sensitive to ciprofloxacin,

imipenem and aztreonam, and resistant to piperacillin/tazobactam, ceftazidime and cefepime.

Fungal biomarkers

Serum β-D-glucan levels measured on Days 2 and 6 were within borderline or non-diagnostic ranges.

Metagenomic cfDNA Analysis

Following the clinical diagnosis of sepsis on Day 3 (see the Definition box), plasma was collected for research-based metagenomic analysis with informed consent (approved by the regional ethics committee; study ID 2020–15535). Plasma was processed through a standardised protocol: centrifugation, cryopreservation and shipment on dry ice to a research facility in the United Kingdom (UK). The plasma sample was processed using a proprietary microfluidic extraction method and sequenced on the MinION Mk1C platform (Oxford Nanopore Technologies) for 2 hours. Details of this approach (iSEP-SEQ) are available in DOI: 10.1101/2024.12.29.24319315. Bioinformatic analysis using the EPI2ME WIMP workflow identified *Klebsiella* (160 reads per million) and *Pseudomonas* (53 reads per million) at the genus level. All other classifications were below interpretative thresholds (<30 RPM), consistent with background signal or contamination. The complete workflow took less than 6 hours. Results were not available in real time but were obtained blindly,

Table 2 Integrated timeline of clinical course, interventions, antimicrobial therapy and microbiological/cfDNA findings during intensive care (ICU) admission

Day	Clinical course and interventions	Key outcomes	Antimicrobials	Microbiology/cfDNA
ICU admission (Day 0)	Alert (GCS 14); pupils equal/reactive. HR 120 bpm, SBP 110 mm Hg, SpO ₂ 98% (3 L/min O ₂ via NC). Abdomen soft, moderately distended, no guarding; drains with minor haemorrhagic output. P/T started intra- and postoperatively.	Initial assessment on transfer to ICU showed moderate abdominal distension, mild tachycardia and adequate oxygenation.	Piperacillin–tazobactam (empiric)	
Day 1	HR ↑150 bpm, hypotension. TTE: hypovolaemia. Worsening abdominal pain. US: small free fluid Douglas pouch. Gelatin-based fluid replacement (Gelafundin, B. Braun). Tremors, shivering, respiratory distress→treated with morphine. IV fluids.	HR partially improved, instability persisted. PE ruled out based on admission imaging.	Piperacillin–tazobactam (continued)	
Day 2	Persistent symptoms. Urology consulted Allergic reaction (pruritus, flushing, wheeze, agitation)→discontinued treatment. Allergic reaction managed with dimetindene, cimetidine, methylprednisolone. Respiratory symptoms treated with salbutamol and ipratropium.	Allergic symptoms resolved. Stable with fluids.	Meropenem+fluconazole (escalated) Meropenem+fluconazole→stopped;	Blood cultures drawn (later negative)
	Second allergic reaction lactate 3.2 mmol/L. Abdominal exam soft, tender, no peritonitis.	Allergic symptoms resolved. Stable with fluids.	Piperacillin–tazobactam re-challenge→stopped; →Ceftriaxone	
Day 3	Respiratory symptoms improved; tachycardia persisted (>130 bpm). Inflammatory markers ↑, PT ↓ to 30%. Lung US: bilateral diaphragmatic elevation, basal atelectasis, small pleural effusion. Furosemide 10 mg.	Fluid balance improved. Metagenomic cfDNA assay identified <i>Klebsiella</i> , <i>Pseudomonas</i> .	Ceftriaxone ongoing (tolerated)	cfDNA: <i>Klebsiella</i> , <i>Pseudomonas</i> (not real time)
Day 4	Dyspnoea→relieved after furosemide 40 mg. CXR: bilateral diaphragmatic elevation, pleural effusions. Inflammatory markers declined.	Symptomatic improvement.	Ceftriaxone ongoing	
Day 5	Nausea, vomiting (treated with dimenhydrinate). Purulent wound drainage collected. −4235 mL negative fluid balance. Abdominal cramps→further urology consult.	GI symptoms controlled.	Ceftriaxone ongoing	Drain cultures remained negative
Day 6	Alert, pain-free. Stable vitals: BP 110/70 mm Hg, HR 100 bpm. Cautious oral diet continued.	Fluctuating infection markers.	Ceftriaxone ongoing	Blood cultures remained negative.
Days 7–9	Persistent elevated infection markers. Blood cultures remained negative.	Clinical improvement; transferred to urology ward in stable condition.	Ceftriaxone completed (7 days total)	Drainage fluid (day 8) and urine cultures (day 9) confirmed cfDNA pathogens: <i>Klebsiella</i> and <i>Pseudomonas</i> .

cfDNA, cell-free DNA; CXR, chest radiograph; GI, gastrointestinal; HR, heart rate; IV, intravenous; NC, nasal cannula; PE, pulmonary embolism; P/T, piperacillin/tazobactam; SBP, systolic blood pressure; SpO₂, peripheral oxygen saturation; TTE, transthoracic echocardiography; US, ultrasound.

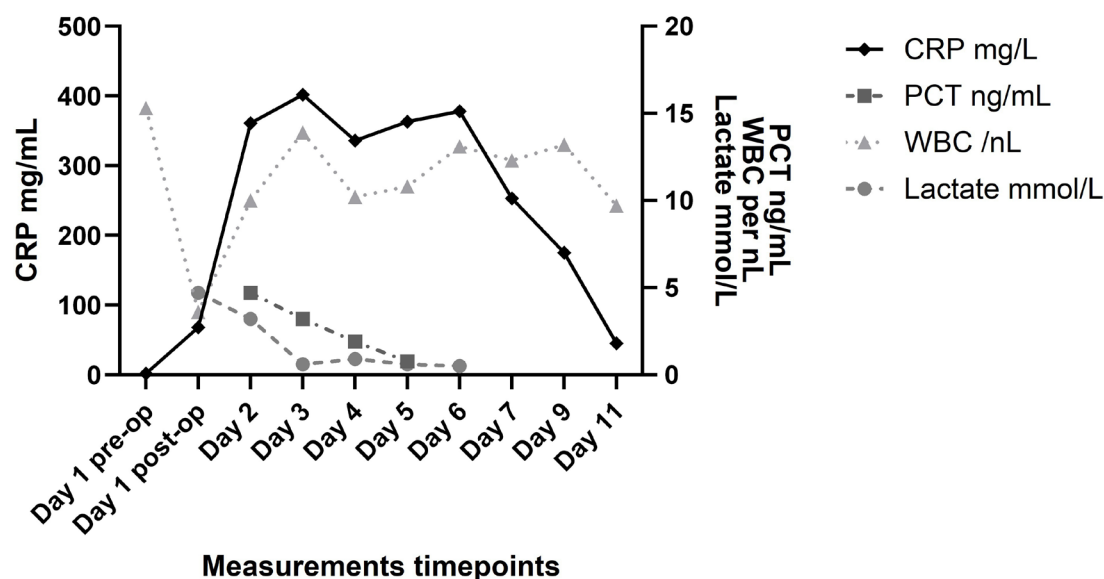


Figure 1 Infectious disease laboratory course: from emergency admission to ICU discharge. CRP, C-reactive protein; ICU, intensive care unit; PCT, procalcitonin; WBC, white blood cell.

shared with the clinical team, and were later found to correlate with culture results from drainage fluid and urine samples.

Interpretation and Limitations

While standard blood cultures remained negative, cfDNA metagenomic sequencing yielded early identification of *Klebsiella* and *Pseudomonas*, later confirmed by conventional cultures of the abdominal drain fluid (table 3). This case highlights both the diagnostic potential of cfDNA-based sequencing and the current limitations related to real-time availability and clinical integration.

DIFFERENTIAL DIAGNOSIS

Differential diagnoses considered during the ICU course included hypovolaemia, postoperative pain-related respiratory compromise, pulmonary embolism, antibiotic-induced hypersensitivity reactions, and sepsis.

Initial tachycardia and echocardiographic findings of kissing ventricles and a collapsible inferior vena cava supported hypovolaemia and prompted aggressive fluid resuscitation. Partial heart rate improvement suggested

volume responsiveness, but persistent tachycardia and respiratory distress indicated an additional process.

Pulmonary embolism was excluded by contrast-enhanced chest CT and normal D-dimer levels. A temporal association between β -lactam administration and respiratory symptoms, with rapid resolution after withdrawal, supported antibiotic-induced hypersensitivity.

Despite these considerations, progressive organ dysfunction, rising inflammatory markers, hyperlactataemia and a Δ SOFA ≥ 2 fulfilled Sepsis-3 criteria, leading to the clinical diagnosis of sepsis despite persistently negative blood cultures.

OUTCOME AND FOLLOW-UP

Following her transfer from the ICU, the patient underwent early mobilisation and respiratory therapy as part of standard postintensive care rehabilitation. The surgical wound healed per primam intentionem, and drainage was removed as planned.

During the remainder of her hospital stay, the patient remained afebrile, and inflammatory markers declined steadily. She completed a 7-day course of intravenous ceftriaxone without adverse effects. She was discharged on Day

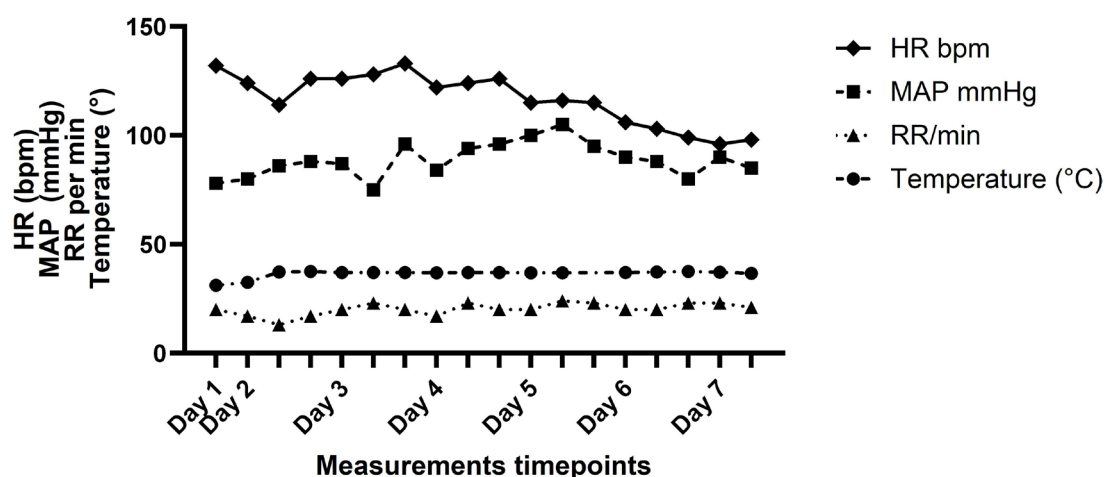


Figure 2 Trends in vital parameter throughout ICU stay. HR, heart rate; ICU, intensive care unit; MAP, mean arterial pressure; RR, respiratory rate.

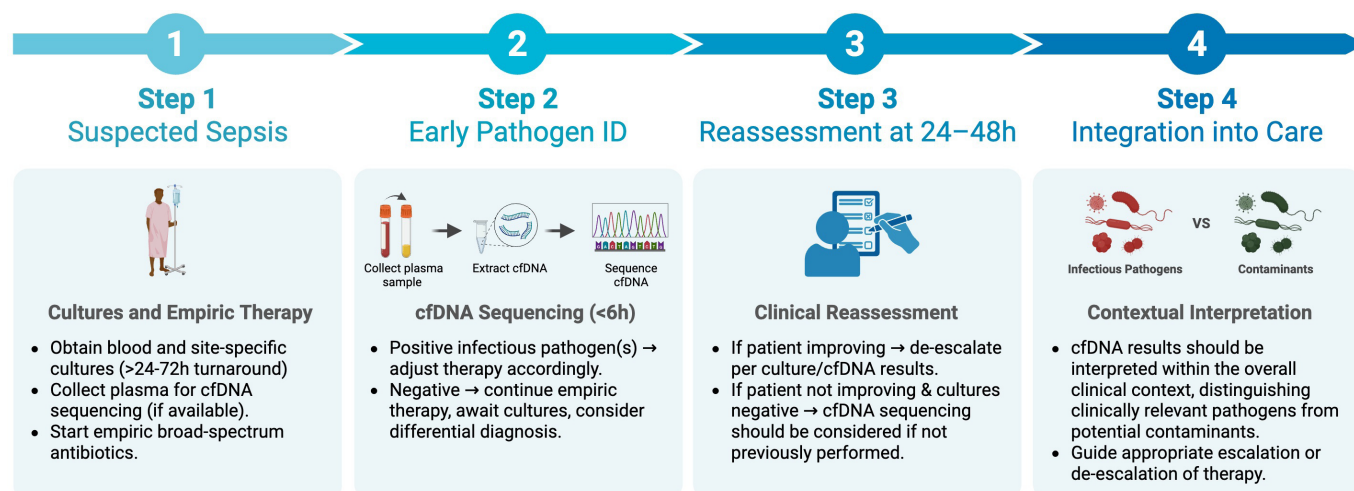


Figure 3 Simplified diagnostic pathway (steps 1–4) for suspected sepsis, integrating cell-free DNA (cfDNA) sequencing with cultures. Step 2 shows detection of microbial cfDNA fragments directly from plasma using metagenomic sequencing, enabling pathogen identification within hours, even when cultures remain negative.

14, in stable condition, with ongoing urinary drainage via the Pouchofix system and a catheter.

At 4-week follow-up, the patient reported no symptoms of infection or complications. Laboratory parameters, including white cell count and C-reactive protein, had normalised. The surgical site remained fully healed, and she had resumed most activities of daily living.

The patient remains under regular outpatient surveillance. At the latest follow-up (15 months after discharge), she was alive, clinically stable and had experienced no recurrence of infection or hospital readmission.

Table 3 Microbiological testing and results throughout hospital stay

	Sampling day			
	Day 2	Day 3	Day 5	Day 7
Blood culture Central venous catheter	Negative result (Day 8)			
Blood culture arterial catheter	Negative result (Day 8)			
Urine (conduit)	Negative result (Day 4)			
Urine (catheter)				Negative result (Day 8)
Urine (pouch)				<i>Pseudomonas aeruginosa</i> (Day 8)
Easy-flow drainage	<i>Klebsiella oxytoca</i> (Day 7)		Negative result (Day 10)	<i>Enterococcus faecium</i> (VRE) (Day 13)
β-D-glucan	Positive result (Day 3)			
Metagenomic plasma analysis		<i>Klebsiella</i> genus (160 RPM) <i>Pseudomonas</i> genus (53 RPM)		

In parenthesis, day at which the results were available.

DISCUSSION

This case underscores the diagnostic and therapeutic complexity of sepsis in patients with altered anatomy following complex urological reconstruction. Structural modifications, such as continent urinary diversions, complicate physiological drainage, sample acquisition and localisation of infection and may delay diagnosis. Similar diagnostic complexities and the need for early imaging in patients with complicated urinary diversions support the findings of this case.^{8,9} Urinary conduit (pouch) rupture should be suspected in patients presenting with acute abdominal pain, reduced catheter output or systemic decompensation, and early imaging is critical for timely surgical management.

While structural complications may be identified promptly with imaging, microbiological confirmation of sepsis-causing pathogens is frequently delayed despite early sampling and empirical therapy.^{10,11} According to Sepsis-3 guidelines, early empirical antibiotic therapy should be initiated within the first hour of recognition, with de-escalation guided by microbiological data.¹ In practice, delayed or absent pathogen identification frequently limits timely de-escalation, as illustrated in this case.

Initial treatment focused on volume resuscitation and empiric antimicrobial therapy with piperacillin–tazobactam. Clinical deterioration prompted escalation to meropenem and fluconazole, while hypersensitivity reactions necessitated an urgent switch to ceftriaxone.¹² Despite ongoing clinical deterioration, standard microbiological diagnostics remained negative for the first 7 days. This is not uncommon in urinary tract infections.^{13,14} Subsequent identification of *Klebsiella oxytoca* and *Pseudomonas aeruginosa* in drainage fluid occurred only after clinical improvement, rendering culture results confirmatory rather than actionable.

A metagenomic microbial cfDNA assay performed on Day 3 samples detected both *Klebsiella* and *Pseudomonas* several days before culture confirmation on drainage fluid. However, as the test was not part of routine clinical diagnostics, results were not used to guide therapy. Published data suggest that plasma metagenomic cfDNA assays can achieve sensitivities of approximately 75%–90% with analytical specificities >99% and clinical specificities of 62%–97% depending on the interpretative thresholds applied and the specific context.^{15–18} Plasma cfDNA sequencing

Table 4 Summary of case reports demonstrating the clinical application of metagenomic next-generation sequencing for early pathogen identification in complex infections

Citation	Clinical scenario	Conventional diagnostics	mNGS findings	NGS method applied	Impact on management
¹²	Elderly female with rheumatoid arthritis and acute respiratory failure on immunosuppression.	Routine microbiological tests (cultures, BALF analyses) negative for pathogens.	Detected <i>P. jirovecii</i> in BALF and serum, confirmed by qPCR.	Plasma and BALF metagenomic NGS (platform not specified).	Confirmation of suspected <i>Pneumocystis pneumonia</i> enabling targeted therapy and supporting clinical decision-making.
¹³	74-year-old female with persistent fever and suspected endocarditis after cardiac surgery.	Negative blood cultures; transesophageal echocardiography suggested vegetation.	Identified <i>Lactobacillus fermentum</i> from plasma microbial cfDNA sequencing.	Plasma microbial cfDNA sequencing (platform not specified).	Enabled etiological diagnosis of culture-negative endocarditis and guided targeted antibiotic therapy.
¹⁴	53-year-old male presenting with symptoms of intracranial infection.	Blood and CSF cultures negative; serology inconclusive.	Detected <i>Lichtheimia ramosa</i> (<i>Mucor</i> species) in CSF.	Unbiased metagenomic next-generation sequencing of CSF (platform not specified).	Prompted antifungal therapy and initial clinical improvement; however overall outcome remained poor.
¹⁵	15-year-old boy with severe pneumonia complicated by systemic infection.	Influenza A detected by PCR from throat swab; cultures inconclusive, bronchoscopy suggested possible alternative pathogens.	Detection of <i>Staphylococcus aureus</i> in BALF and peripheral blood; MRSA later confirmed by culture.	Unbiased metagenomic next-generation sequencing of BALF and peripheral blood (platform not specified).	Enabled antimicrobial escalation (initiation of anti-MRSA therapy and discontinuation of antifungals).
¹⁶	53-year-old man with persistent fever of unknown origin.	Routine microbiological tests negative.	Detection of <i>Coxiella burnetii</i> DNA in blood.	Metagenomic next-generation sequencing of blood (platform not specified).	Led to diagnosis of Q fever and initiation of targeted antibiotic therapy.

has demonstrated superior detection of Gram-negative pathogens compared with conventional blood cultures, including *Enterobacteriales* and *Pseudomonas*.¹⁹ In a recent prospective study, plasma cfDNA mNGS detected 78% of Gram-negative rods in blood-stream infections, surpassing traditional culture performance.²⁰ Several medical case reports have reported the use of cfDNA sequencing for the detection of pathogens. Table 4 provides an overview of cases in which cfDNA sequencing revealed pathogens not identified by initial culture-based methods.^{21–25} Our case complements other published reports to support the notion that metagenomics could improve outcomes by enabling earlier targeted therapy and subsequent de-escalation and reduction of antimicrobial exposure. *Klebsiella* and *Pseudomonas* are common Gram-negative uropathogens, and their detection by cfDNA is biologically expected in severe postoperative infection. This case is distinguished from previous reports by the use of real-time nanopore sequencing for pathogen detection directly from cfDNA extracted by a novel sample preparation platform, CNASafe. Most existing cases have relied on large-scale laboratory infrastructure, whereas this approach demonstrates the feasibility of decentralised diagnostics. In addition, this case uniquely demonstrates the detection of uropathogens from plasma cfDNA sequencing.

Compared with standard cultures, cfDNA sequencing offers earlier, culture-independent pathogen detection, but cannot distinguish viable from non-viable organisms and does not reliably localise the site of infection. In addition, limitations of cfDNA sequencing include susceptibility to contamination, variable reporting thresholds and heterogeneity in laboratory workflows and bioinformatic pipelines.^{26–27} Accordingly, cfDNA assays should be regarded as complementary to conventional diagnostics in diagnostically complex or culture-negative cases. The metagenomic cfDNA assay used in this case was performed within a research framework. The Oxford Nanopore Technologies MinION Mk1C platform was selected for its on-demand sequencing capability and potential applicability in near-patient settings. Although the complete analytical workflow required less than 6 hours, results were not available in real time and did not inform clinical decision-making.

It should be noted that the Oxford Nanopore workflow offers technical capability for concurrent AMR gene detection.²⁸ While our analysis focused on species-level identification, incorporating resistance profiling would provide a more comprehensive diagnostic tool. This is particularly relevant in regions with higher prevalence of AmpC or carbapenemase-mediated resistance,^{29–30} expanding the potential for rapid pathogen and resistance profiling in critical settings.

Clinical improvement was multifactorial, driven by surgical source control, empirical antimicrobial therapy and supportive intensive care. The cfDNA findings correlated with abdominal drainage fluid and urine culture results, which identified causative pathogens. The added value of this case lies in the diagnostic context and timing: cfDNA sequencing was applied in a patient with complex postoperative anatomy and persistently negative blood cultures, with pathogen detection preceding culture confirmation by several days.

Current European Association of Urology guidelines recommend empirical broad-spectrum therapy for urosepsis with subsequent de-escalation based on microbiological results.³¹ This case illustrates the diagnostic gap created when culture results are delayed or absent. These developments align with broader public health initiatives, such as the UK metagenomic pathogen surveillance programme.²⁶ Until supported by prospective validation studies and adopted into clinical guidelines, metagenomic cfDNA assays should be regarded as valuable adjuncts rather than replacements for conventional diagnostics in sepsis care.

Our case complements other published reports describing delayed pathogen detection in clinically altered patients and supports the notion that molecular diagnostics could improve outcomes by enabling earlier targeted therapy and subsequent de-escalation and reduction of antimicrobial exposure. Table 4 provides an overview of cases in which cfDNA sequencing revealed pathogens not identified by initial culture-based methods.^{21–25}

Learning points

- Sepsis in multimorbid patients with surgically altered anatomy may present atypically and complicate early diagnosis, especially when conventional microbiological sampling is challenging or delayed.
- Empiric broad-spectrum antibiotic therapy remains essential in early sepsis management, but hypersensitivity reactions may significantly limit therapeutic options, requiring close clinical monitoring and adaptability.
- Conventional culture-based diagnostics often fail to provide timely pathogen identification in complex cases, limiting their utility for early targeted antimicrobial therapy.
- Metagenomic cell-free DNA (cfDNA) sequencing can detect pathogens rapidly and independently of viable cultures—offering a promising adjunct in time-sensitive or diagnostically unclear infections.
- While cfDNA sequencing was not applied in real time in this case, this study highlights its potential as a near-patient adjunctive diagnostic tool within future clinical workflows, reducing reliance on central laboratory infrastructure and improving pathogen-specific decision-making in sepsis care.

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Competing interests MK-K is inventor on several patents owned by Heriot-Watt University, which cover the microfluidic cfDNA extraction platform, CNASafe, mentioned in this study. RSK and MK-K are co-founder are co-founders of Natantis, an early-stage spinout from Heriot-Watt University developing CNASafe technology for cfDNA extraction and its application to pathogen identification using metagenomic sequencing. CNASafe was used for cfDNA extraction in this study. Natantis is not yet a trading entity, and no commercial or financial interests influenced the design, analysis, or reporting of this work.

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Case reports provide a valuable learning resource for the scientific community and can indicate areas of interest for future research. They should not be used in isolation to guide treatment choices or public health policy.

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